

Loveland, Morrissey & Seaborg

Modern Nuclear Chemistry (2nd ed., 2017)

Chapter 8 — Beta Decay

Section 8.3: Derivation of the Spectral Shape

Refresher Notes (aligned with the textbook)

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Summer 2026

This is the heart of the chapter — Fermi’s quantum-mechanical derivation of the β -spectrum shape. The pace is deliberately slow, the algebra is broken into small steps, and notation matches Loveland exactly: weak-interaction strength g , electron kinetic energy T_e , endpoint Q , and Planck’s constant h (rather than $\hbar$) in the phase-space counting. We end where the textbook ends: at Eq. 8.21, the Coulomb-corrected spectrum allowing for a nonzero neutrino mass. The Fermi integral $f(Z_d, Q)$ and ft values appear later, in §8.5.

0. Background you need fresh in your head

0.1 Fermi’s golden rule (the main hammer)

For a small, time-independent perturbation V_p that drives transitions from a single initial state Ψ_i to a continuum of final states Ψ_f , time-dependent perturbation theory gives the transition rate

$$\lambda = \frac{2\pi}{\hbar} \left| \int \Psi_f^* V_p \Psi_i d\tau \right|^2 \rho(E_f), \quad (1)$$

where $\rho(E_f) = dn/dE$ is the density of available final states (per unit final energy). This is Loveland’s Eq. 8.8.

Two mental compartments:

- **Dynamics** live inside the matrix element $\int \Psi_f^* V_p \Psi_i d\tau$.
- **Counting / kinematics** live inside $\rho(E_f)$.

All of §8.3 evaluates these two things and multiplies them.

0.2 Density of momentum states (semiclassical)

For a free particle of momentum $\mathbf{p}$ confined to a normalization volume V , the number of plane-wave states with momentum magnitude in $(p, p + dp)$ is

$$dn = \frac{V 4\pi p^2 dp}{h^3}. \quad (2)$$

The denominator h^3 is the “volume of one cell in phase space” (quantum mechanics says one state per cell). This is the only nontrivial counting in the derivation.

0.3 Plane-wave normalization

In a box of volume V , plane-wave wavefunctions are normalized as

$$\phi(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}}. \quad (3)$$

Loveland writes the electron and neutrino as such radial plane waves (Eqs. 8.14–8.15). The V factors will cancel exactly against V ’s in the density of states; the answer is finite and V -independent.

0.4 Where in the chapter we are

§8.1 named the three modes; §8.2 introduced the neutrino; §8.3 now *computes* what the spectrum should look like. Once we have the spectrum, §8.4 (Kurie plots) gives a graphical way to read off Q and test the assumptions, and §8.5 integrates the spectrum to obtain a total rate and the famous $\log ft$ classification.

1 Why a quantum-mechanical theory at all?

The textbook opens with two observations that motivate Fermi’s whole program:

1. Beta decay obeys first-order kinetics — it is governed by a single decay constant. So we want λ .
2. Half-lives span 18 orders of magnitude (a few ms to $\sim 10^{16}$ yr), reminiscent of α decay. Nuclear structure (parent state, daughter levels) must therefore play a central role.

Key Idea

The key conceptual analogy: β decay is like spontaneous photon emission from an excited atom. Both processes start from a single quantum state of well-defined energy and *create* new particles. Fermi imported the formalism that already worked for photon emission — first-order time-dependent perturbation theory, i.e., the golden rule — and applied it to a new “weak” interaction whose strength constant must be determined from experiment.

The new strength constant is denoted g (Loveland’s notation; in modern notation G_F). Its empirical value is

$$g \approx 0.88 \times 10^{-4} \text{ MeV}\cdot\text{fm}^3, \quad (4)$$

about 10^{-3} of the electromagnetic coupling $e^2/(4\pi\epsilon_0)$. That single numerical fact is the deepest reason β -decay half-lives are typically $\gg$ atomic-transition lifetimes.

2 Setting up the matrix element

2.1 The initial and final wavefunctions

The full state of the system before and after the decay is:

$$\Psi_i = \phi_{\text{gs}}(Z^A \text{X}) \quad (\text{parent ground state only}) \quad (5)$$

$$\Psi_f^* = \phi_j^*(Z^A \text{Y}) \phi^*(e) \phi^*(\nu) \quad (\text{daughter in state } j, \text{ electron, neutrino}). \quad (6)$$

The final wavefunction is a product because the three outgoing pieces are non-interacting after the decay (we treat the daughter as inert and the leptons as free).

2.2 Plane waves at the nuclear point

Substitute the box-normalized plane waves

$$\phi_e(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}_e \cdot \mathbf{r}}, \quad \phi_\nu(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}_\nu \cdot \mathbf{r}}, \quad (7)$$

into the matrix element. Across the nuclear volume ($r \sim R \sim \text{a few fm}$), the lepton momenta times r are small ($kr \ll 1$ for $\sim \text{MeV}$ leptons), so we can Taylor-expand:

$$e^{i\mathbf{k} \cdot \mathbf{r}} = 1 + i\mathbf{k} \cdot \mathbf{r} + \dots \approx 1. \quad (8)$$

Keeping only the leading “1” is what makes a transition *allowed*; keeping the next term gives *first-forbidden* decays (§8.6), which carry one unit of orbital angular momentum.

Common Pitfall

Don’t lose track of why kr is small. For a 1 MeV electron, $\hbar/p_e \approx 200 \text{ fm}$, while the nuclear radius is a few fm. So $k_e r = p_e r / \hbar \sim 0.02$. Truncating the expansion is genuinely a leading-order approximation, not a hand-wave.

In this allowed approximation, $\phi_e(0) = \phi_\nu(0) = 1/\sqrt{V}$ (Loveland’s Eq. 8.17), and the weak-interaction matrix element factorizes cleanly:

$$\int \Psi_f^* V_p \Psi_i d\tau \approx \frac{g}{V} M_{fi}, \quad M_{fi} \equiv \int \phi_j^* O_\beta \phi_{\text{gs}} d\tau_N, \quad (9)$$

where M_{fi} is the *nuclear* matrix element. It contains all the information about the overlap of the parent and daughter nuclear wavefunctions, with O_β the operator that converts a neutron into a proton (or vice versa). M_{fi} is what one needs the shell model (or another structure calculation) to evaluate. The square is

$$\left| \int \Psi_f^* V_p \Psi_i d\tau \right|^2 = \frac{g^2}{V^2} |M_{fi}|^2. \quad (10)$$

3 The density of final states

3.1 Counting two-lepton phase space

The final state contains *two* free particles — the electron and the neutrino. The recoiling daughter nucleus is heavy and carries negligible kinetic energy, so we ignore its phase space. Apply Eq. (2) to each lepton independently:

$$dn_e = \frac{V 4\pi p_e^2 dp_e}{h^3}, \quad (11)$$

$$dn_\nu = \frac{V 4\pi p_\nu^2 dp_\nu}{h^3}. \quad (12)$$

The total number of final states with the leptons in those momentum shells is the product (Loveland's Eq. 8.9):

$$dn = \frac{16\pi^2 V^2 p_e^2 p_\nu^2 dp_e dp_\nu}{h^6}. \quad (13)$$

3.2 Eliminating the neutrino momentum

Energy conservation locks the neutrino energy once the electron energy is fixed:

$$T_\nu = Q - T_e. \quad (14)$$

Assume the neutrino is massless, so $T_\nu = p_\nu c$ (Loveland's Eq. 8.10):

$$p_\nu = \frac{Q - T_e}{c}, \quad dp_\nu = \frac{dQ}{c}. \quad (15)$$

Substitute into Eq. (13) and read off the *differential* density of final states (Loveland's Eq. 8.13):

$$\frac{dn}{dQ} = \frac{16\pi^2 V^2}{h^6 c^3} (Q - T_e)^2 p_e^2 dp_e. \quad (16)$$

Common Pitfall

What does d/dQ mean here? As Loveland gently warns, Q is not a variable being differentiated against in the usual sense — Q is fixed for a given decay. The notation captures how the count of final states changes with the *available energy* Q at fixed electron-momentum slice. Operationally, you should think of Eq. (16) as “ $\rho(E_f)$ for the slice of phase space with electron momentum in $(p_e, p_e + dp_e)$.”

4 Putting it together: the bare spectrum

Combine: golden rule (1), matrix-element square (10), and density of states (16). Watch the V^2 's cancel:

Derivation

$$\begin{aligned} \lambda(p_e) dp_e &= \frac{2\pi}{\hbar} \frac{g^2}{V^2} |M_{fi}|^2 \frac{16\pi^2 V^2}{h^6 c^3} (Q - T_e)^2 p_e^2 dp_e \\ &= \frac{|M_{fi}|^2 g^2}{2\pi^3 \hbar^7 c^3} (Q - T_e)^2 p_e^2 dp_e. \end{aligned}$$

The constants are reorganized using $h = 2\pi\hbar$, so $h^6 = 64\pi^6 \hbar^6$, and the $32\pi^3$ in numerator/denominator collapse to give the familiar coefficient.

This is Loveland's Eq. 8.18:

$$\boxed{\lambda(p_e) dp_e = \frac{|M_{fi}|^2 g^2}{2\pi^3 \hbar^7 c^3} (Q - T_e)^2 p_e^2 dp_e.} \quad (17)$$

4.1 Reading the formula — “the statistical factor”

Pull the constants out and look at what’s left:

$$\lambda(p_e) dp_e = (\text{constants}) (Q - T_e)^2 p_e^2 dp_e. \quad (18)$$

This kinematic shape $(Q - T_e)^2 p_e^2$ is what Loveland calls the **statistical (or phase-space) factor**. Each piece has a clear meaning:

- p_e^2 — electron phase space; vanishes at $p_e \rightarrow 0$ (rising edge of the spectrum).
- $(Q - T_e)^2$ — neutrino phase space; vanishes *quadratically* at the endpoint $T_e \rightarrow Q$ (falling tail).
- The peak sits between, and the **width of the spectrum is set by Q** , which is why Q is the most fundamental beta-decay quantity.

Key Idea

The continuous β spectrum — the puzzle of §8.2 — here drops out automatically as the convolution of two phase-space factors. There is no “missing energy”; the spectrum is just what you get when you share Q between two leptons. Fermi’s theory *predicts* the shape, not just accommodates it.

The shape is Loveland’s Figure 8.1: $N(p)$ rises like p^2 at low momentum and falls parabolically toward $p_{\max}$; $N(T)$ is similar but with the energy variable.

5 The Coulomb correction: Fermi function $F(Z_d, p_e)$

The plane-wave step in Eq. (7) ignored the Coulomb field of the daughter nucleus. That is the biggest single approximation in the derivation, and it shifts the spectrum:

- For β^- decay (daughter $Z_d = Z + 1$, attractive to the electron), low-momentum electrons are **enhanced** — they spend more time near the nucleus.
- For β^+ decay (daughter $Z_d = Z - 1$, repulsive to the positron), low-momentum positrons are **suppressed**.

Fermi treated this by replacing the lepton plane waves with Coulomb-distorted wavefunctions evaluated at the nuclear surface. The resulting multiplicative correction is the **Fermi function** $F(Z_d, p_e)$:

$$\lambda(p_e) dp_e = (\text{constants}) F(Z_d, p_e) (Q - T_e)^2 p_e^2 dp_e. \quad (19)$$

This is Loveland’s Eq. 8.20. The textbook illustrates the effect with ^{64}Cu , which decays by both β^- and β^+ to neighboring even–even daughters with similar Q values, so the Coulomb distortion shows up as a clean asymmetry between the two spectra (Figure 8.2).

Common Pitfall

$F(Z_d, p_e)$ depends on the *daughter* charge, not the parent’s. Easy to slip — write down whether the decay is β^- , β^+ , or EC first, then identify Z_d .

6 Allowing a nonzero neutrino mass

Removing the $m_\nu = 0$ assumption changes the neutrino phase space factor. Energy conservation now reads $E_\nu^2 = (p_\nu c)^2 + m_\nu^2 c^4$ with $E_\nu = Q - T_e$, so

$$p_\nu = \frac{1}{c} \sqrt{(Q - T_e)^2 - m_\nu^2 c^4}. \quad (20)$$

Walking this back through the density-of-states calculation produces an extra factor:

$$\lambda(p_e) dp_e = \frac{|M_{fi}|^2}{2\pi^3 \hbar^7 c^3} F(Z_d, p_e) (Q - T_e)^2 \left[1 - \frac{m_\nu^2 c^4}{(Q - T_e)^2} \right]^{1/2} p_e^2 dp_e. \quad (21)$$

This is Loveland's Eq. 8.21 (citing Heyde 2005), and it is the **endpoint** of §8.3.

Key Idea

The neutrino-mass correction matters *only* very near the spectrum endpoint, where $(Q - T_e)$ is comparable to $m_\nu c^2$. Far from the endpoint, the bracketed factor $\rightarrow 1$ and Eq. (19) is recovered. This is precisely why precision measurements of the endpoint of tritium β decay ($Q = 18.6$ keV, very small) provide the best laboratory bound on m_{ν_e} . KATRIN currently sets $m_{\nu_e} \lesssim 0.8$ eV.

7 Summary — what §8.3 actually delivers

- Fermi's golden rule (Eq. 8.8) is the engine.
- Initial state Ψ_i = parent g.s.; final state Ψ_f = daughter in state $j \times$ electron plane wave $\times$ neutrino plane wave.
- Long-wavelength (allowed) approximation: $e^{i\mathbf{k}\cdot\mathbf{r}} \approx 1$ across the nuclear volume. Matrix element collapses to $g M_{fi}/V$.
- Density of final states from two-lepton phase space: $dn \propto V^2 p_e^2 p_\nu^2 dp_e dp_\nu / h^6$; massless- ν kinematics turns this into $dn/dQ \propto p_e^2 (Q - T_e)^2 / c^3$ (Eq. 8.13).
- Combining these in the golden rule gives Eq. 8.18, $\lambda(p_e) dp_e \propto |M_{fi}|^2 g^2 p_e^2 (Q - T_e)^2 dp_e$. The volume factors cancel exactly.
- The Coulomb correction multiplies by $F(Z_d, p_e)$ (Eq. 8.20), enhancing β^- at low energy and suppressing β^+ at low energy.
- Allowing $m_\nu > 0$ adds the bracket $[1 - m_\nu^2 c^4 / (Q - T_e)^2]^{1/2}$ (Eq. 8.21), with observable effect only near the endpoint.

What's coming next. §8.4 takes Eq. 8.20, divides by $p_e^2 F(Z_d, p_e)$, takes the square root, and plots vs. T_e — a *Kurie plot*. If the matrix element is energy-independent (allowed transition, no forbidden corrections), the plot is a straight line whose intercept is Q . §8.5 then integrates Eq. 8.20 over p_e to get the total rate, defines the dimensionless Fermi integral $f(Z_d, Q)$, and finally introduces the comparative half-life $ft_{1/2}$ that lets us classify transitions independent of Q .